

Sophia Vaughan



Postdoc fellow

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RESEARCH INTERESTS

- The characterisation of exoplanet atmospheres at high spectral resolution.
- Biosignatures, technosignatures and the search for extraterrestrial life.

RESEARCH EXPERIENCE

2024-	Max Planck Institute for Astronomy, APEX prize fellowship, Supervisor: Prof. Laura Kreidberg High spectral resolution characterisation of exoplanets in reflected light.
2020-2024	University of Oxford, Astrophysics DPhil, Supervisor: Prof. Jayne Birkby Developed high spectral resolution techniques for characterising exoplanets in reflected light.
2019-2020	University of Oxford, Master's Project, Supervisor: Prof. Suzanne Aigrain Worked on using Gaussian Processes to model stellar spectra and measure stellar radial velocities.
2019	Joint Institute of VLBI - ERIC (JIVE), Supervisor: Prof. Huib Jan van Langevelde Studied inverse phase referencing of MASERs using CASA (Common Astronomy Software Applications).
2018	University of Oxford Astrophysics Summer Project, Supervisor: Dr Helen Spiers Worked with the Zooniverse team on public outreach projects.

EDUCATION

2020-2024	DPhil Astrophysics University of Oxford, St John's College Thesis: <i>Reflected light from exoplanet atmospheres: towards rocky worlds</i> Supervisors: Prof Jayne Birkby and Prof Raymond Pierrehumbert
2016-2020	Master of Physics, 1st Class Honours, 8th highest score in the cohort University of Oxford, St John's College Masters Project: <i>Detection of Transiting Exoplanets with the TESS Space Mission</i> Supervisor: Prof Suzanne Aigrain

SUPERVISING AND TEACHING EXPERIENCE

Supervisor to Master student, Max Planck Institute for Astronomy, Heidelberg Germany	2026
Supervisor to internship student, Max Planck Institute for Astronomy, Heidelberg Germany	2026
Waynflete Studies Programme tutor, Magdalen College School, Oxford UK	2023
Atmospheric Physics Marker, Department of Physics, University of Oxford, UK	2022-2024
Astrophysics TA, Department of Physics, University of Oxford, UK	2022
Mentor to Masters Student, Department of Physics, University of Oxford, UK	2021-2022

CONTRIBUTIONS TO ACADEMIA

<u>Peer Review</u>	
Monthly Notices of the Royal Astronomical Society	2024-
Astronomy and Astrophysics	2025-2026
<u>Major International Collaborations and Consortia</u>	
HARMONI consortium	2023-
<u>Organisation of Scientific Meetings</u>	
SOC of breakout session at UK National Astronomy Meeting	Jul 2022

GRANTS, HONOURS AND AWARDS

RAS Thesis Prize , Royal Astronomical Society, Burlington House, London, UK	2024	£1000
Gibbs Prize , Department of Physics, University of Oxford, UK	2019	£200
Gibbs Prize , Department of Physics, University of Oxford, UK	2018	£150
Casberd Scholarship , St. John's College, University of Oxford, UK	2017-2020	£300/yr

CONFERENCES

MAVIS 2026	Garching, Germany	Highlight Talk	Sep 2026
HoRS3s	Granada, Spain	Contributed Talk	Jul 2026
Characterizing Habitable Exoplanets with Interdisciplinary Expertise	Leiden, Netherlands	Invited Workshop	May 2026
Planetary formation and Exoplanets in the ELT era	Garching, Germany	Contributed Talk	Nov 2025
Exoclimates VII	Montreal, Canada	Contributed Talk	Jul 2025
Two HoRSEs	Berlin, Germany	Contributed Talk	Jul 2024
Exoplanets Atmospheres to Architectures	Washington, US	Contributed Talk	Sep 2023
Exoplanets by the Lake	Munich, Germany	Contributed Talk	Aug 2023
Science with the Habitable Worlds Observatory and Beyond	Baltimore, US	Contributed Talk	Jul 2023
National Astronomy Meeting	Cardiff, UK	Contributed Talk	Jul 2023
Optimal Exoplanet Imagers	Leiden, Netherlands	Invited Workshop	Feb 2023
UKEXOM	Edinburgh, Scotland	Highlight Talk	Sept 2022
National Astronomy Meeting	Warwick, UK	Contributed Talk	Jul 2022
Rocky Worlds II	Oxford, UK	Contributed Talk	Jul 2022

PRESS RELEASES AND OUTREACH

Outreach	<i>Girls Day</i> , Heidelberg, Germany	Volunteer	Apr 2026
Press	Commentary: Sophia Vaughan on searching for an exo-rainbow	Commentary	Jul 2023
Outreach	<i>Into the cosmos</i> , Oxford, UK	Volunteer	Jan 2023
Outreach	<i>Passport to the Solar System</i> , Oxford, UK	Volunteer	Jun 2021

TELESCOPE OBSERVATIONS

Gemini/GHOST	sPI	2026A FT	7 hours	SM	<i>Determining the altitude of HD189733b’s clouds/hazes using reflected light spectroscopy with GHOST</i>
VLT(16.4m)/ESPRESSO	PI	2023B	2.5 nights	VM/dVM	<i>Mirror in the Desert: a high resolution reflection spectrum of the unusual ultra hot Neptune LTT-9779 b</i>
VLT(8.2m)/CRIRES+	Co-I	2023B	6.6 hours		<i>A Desert Island: Probing the Origins of the Desert with Atmospheric Transmission of a Hot Neptune</i>
VLT(8.2m)/ESPRESSO	Co-I	2022B	30 hours		<i>Through the Looking Glass: a high resolution ESPRESSO survey of exoplanet reflection spectra</i>
VLT(8.2m)/CRIRES+	Co-I	2022A	8 hours	dVM	<i>Revealing an atmosphere shrouded in mystery with high-resolution spectroscopy</i>

DATA ANALYSIS EXPERIENCE

VLT(8.2m)/ESPRESSO	Data	Experience in analysing high-resolution spectroscopy observations of exoplanets in reflected light and of the unique challenges this poses.
VLT(16.4m)/ESPRESSO	Data	Developed analysis of high-resolution spectroscopy observations of exoplanets in reflected light with the 4UT mode and the differences between this and single UT mode.
VLT(8.2m)/CRIRES+	Simulation	Created simulated high-resolution spectroscopy observations of exoplanets (transit and emission) and analysed them to verify the observational requirements for observing proposals.
VLT(8.2m)/ESPRESSO	Simulation	Developed a high-resolution simulation of ESPRESSO observations of exoplanets (reflection) and analysed them to verify the observational requirements for observing proposals.
ELT(39m)/HARMONI	Simulation	Experience in simulating spatially resolved observations of exoplanets (reflection) and analysing them using the ‘molecule mapping’ method to determine science cases for the present instrument design.

LANGUAGES

Computing: Python, LaTeX and bash. Basic knowledge of: AIPS and Matlab
Spoken: English (native), Norwegian (self-taught ~A2)

First Author:

- [1] Sophia R. Vaughan, Jayne L. Birkby, Natasha E. Batalha, Luke T. Parker, Haochuan Yu, Julia V. Seidel, Michael Radica, Jake Taylor, Laura Kreidberg, Vivien Parmentier, Sergio Hoyer, James S. Jenkins, Annabella Meech, Ricardo Ramírez Reyes, and Lennart van Sluijs. No TiO detected in the hot-Neptune-desert planet LTT-9779 b in reflected light at high spectral resolution. *A&A*, 705:A27, January 2026.
- [2] Sophia R. Vaughan, Jayne L. Birkby, Niranjana Thatte, Alexis Carlotti, Mathis Houllé, Miguel Pereira-Santaella, Fraser Clarke, Arthur Vigan, Zifan Lin, and Lisa Kaltenegger. Behind the mask: can HARMONI@ELT detect biosignatures in the reflected light of Proxima b? *MNRAS*, 528(2):3509–3522, February 2024.
- [3] Sophia R. Vaughan, Timothy D. Gebhard, Kimberly Bott, Sarah L. Casewell, Nicolas B. Cowan, David S. Doelman, Matthew Kenworthy, Johan Mazoyer, Maxwell A. Millar-Blanchaer, Victor J. H. Trees, Daphne M. Stam, Olivier Absil, Lisa Altinier, Pierre Baudoz, Ruslan Belikov, Alexis Bidot, Jayne L. Birkby, Markus J. Bonse, Bernhard Brandl, Alexis Carlotti, Elodie Choquet, Dirk van Dam, Niyati Desai, Kevin Fogarty, J. Fowler, Kyle van Gorkom, Yann Gutierrez, Olivier Guyon, Sebastiaan Y. Haffert, Olivier Herscovici-Schiller, Adrien Hours, Roser Juanola-Parramon, Evangelia Kleisioti, Lorenzo König, Maaïke van Kooten, Mariya Krasteva, Iva Laginja, Rico Landman, Lucie Leboulleux, David Mouillet, Mamadou N'Diaye, Emiel H. Por, Laurent Pueyo, and Frans Snik. Chasing rainbows and ocean glints: Inner working angle constraints for the Habitable Worlds Observatory. *MNRAS*, 524(4):5477–5485, October 2023.
- [4] Sophia R. Vaughan and Laura Kreidberg. Rotational broadening of exoplanet spectra in arbitrarily oriented systems: Application to reflected light. *arXiv e-prints*, page arXiv:2609.05619, September 2026.

Co-author:

- [5] Xander Byrne, Claire Marie Guimond, Amy Bonsor, Haiyang S. Wang, **Vaughan, Sophia R.**, and James G. Rogers. The Barnard's Star Planetary System: Stability, Composition, and Evolution of Four Sub-Earth Exoplanets. *arXiv e-prints*, page arXiv:2606.23310, June 2026.
- [6] K. Angelique Kahle, Paul Mollière, Laura Kreidberg, Bertram Bitsch, Mara Attia, Silke S. Dainese, Nicholas Storm, Daniel Valentine, Thomas M. Evans-Soma, H. J. Hoeijmakers, Stefan Pelletier, Ludmila Carone, Cyril Gapp, Yoav Rotman, **Vaughan, Sophia R.**, D. A. Christie, Louis-Philippe Coulombe, Christiane Helling, and Thomas Henning. The panchromatic JWST dayside spectrum of WASP-121 b reveals a refractory-rich formation. *arXiv e-prints*, page arXiv:2606.21855, June 2026.
- [7] Luke T. Parker, Jayne L. Birkby, Siddharth Gandhi, Vivien Parmentier, Vatsal Panwar, Matteo Brogi, and **Vaughan, Sophia R.** The CRIMSON survey I: super-stellar SiO in the directly imaged companion TWA 5 B from high-resolution M-band spectroscopy. *MNRAS*, June 2026.
- [8] Luke T. Parker, João M. Mendonça, Hannah Diamond-Lowe, Jayne L. Birkby, Annabella Meech, **Vaughan, Sophia R.**, Matteo Brogi, Chloe Fisher, Lars A. Buchhave, Aaron Bello-Arufe, Laura Kreidberg, and Jason Dittmann. Limits on the atmospheric metallicity and aerosols of the sub-Neptune GJ 3090 b from high-resolution CRRES+ spectroscopy. *MNRAS*, 538(4):3263–3283, April 2025.
- [9] Luke T. Parker, Jayne L. Birkby, Rico Landman, Joost P. Wardenier, Mitchell E. Young, **Vaughan, Sophia R.**, Lennart van Sluijs, Matteo Brogi, Vivien Parmentier, and Michael R. Line. Into the red: an M-band study of the chemistry and rotation of β Pictoris b at high spectral resolution. *MNRAS*, 531(2):2356–2378, June 2024.
- [10] J. Fowler, Sebastiaan Y. Haffert, Maaïke A. M. van Kooten, Rico Landman, Alexis Bidot, Adrien Hours, Mamadou N'Diaye, Olivier Absil, Lisa Altinier, Pierre Baudoz, Ruslan Belikov, Markus Johannes Bonse, Kimberly Bott, Bernhard Brandl, Alexis Carlotti, Sarah L. Casewell, Elodie Choquet, Nicolas B. Cowan, Niyati Desai, David Doelman, Kevin Fogarty, Timothy D. Gebhard, Yann Gutierrez, Olivier Guyon, Olivier Herscovici-Schiller, Roser Juanola-Parramon, Matthew Kenworthy, Elina Kleisioti, Lorenzo König, Mariya Krasteva, Iva Laginja, Lucie Leboulleux, Johan Mazoyer, Maxwell A. Millar-Blanchaer, David Mouillet, Emiel Por, Laurent Pueyo, Frans Snik, Dirk van Dam, Kyle van Gorkom, and **Vaughan, Sophia R.** Visible extreme adaptive optics on extremely large telescopes: Towards detecting oxygen in Proxima Centauri b and analogs. *arXiv e-prints*, page arXiv:2309.00725, September 2023.
- [11] Niyati Desai, Lorenzo König, Emiel Por, Roser Juanola-Parramon, Ruslan Belikov, Iva Laginja, Olivier Guyon, Laurent Pueyo, Kevin Fogarty, Olivier Absil, Lisa Altinier, Pierre Baudoz, Alexis Bidot, Markus Johannes Bonse, Kimberly Bott, Bernhard Brandl, Alexis Carlotti, Sarah L. Casewell, Elodie Choquet, Nicolas B. Cowan, David Doelman, J. Fowler, Timothy D. Gebhard, Yann Gutierrez, Sebastiaan Y. Haffert, Olivier Herscovici-Schiller, Adrien Hours, Matthew Kenworthy, Elina Kleisioti, Mariya Krasteva, Rico Landman, Lucie Leboulleux, Johan Mazoyer, Maxwell A. Millar-Blanchaer, David Mouillet, Mamadou N'Diaye, Frans Snik, Dirk van Dam, Kyle van Gorkom, Maaïke van Kooten, and **Vaughan, Sophia R.** Integrated photonic-based coronagraphic systems for future space telescopes. *arXiv e-prints*, page arXiv:2309.04925, September 2023.
- [12] N. L. Eisner, O. Barragán, C. Lintott, S. Aigrain, B. Nicholson, T. S. Boyajian, S. Howell, C. Johnston, B. Lakeland, G. Miller, A. McMaster, H. Parviainen, E. J. Safron, M. E. Schwamb, L. Trouille, **Vaughan, S.**, N. Zicher, C. Allen, S. Allen, M. Bouslog, C. Johnson, M. N. Simon, Z. Wolfenbarger, E. M. L. Baeten, D. M. Bundy, and T. Hoffman. Planet Hunters TESS II: findings from the first two years of TESS. *MNRAS*, 501(4):4669–4690, March 2021.